

JAYA VERMA

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PROFESSIONAL SUMMARY

AI Engineer who ships production LLM systems end-to-end - data pipelines, agentic multi-agent architectures, and Retrieval-Augmented Generation (RAG) i.e. serving ~190 clients. Known for aggressive cost and prompt optimization (core production prompts cut ~88%, inference at ~\$0.01 per prospect) and a published NLP paper (JETIR 2023).

TECHNICAL SKILLS

LLM & GenAI: Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Agentic AI, Multi-Agent Systems, Prompt Engineering, Fine-Tuning, Embeddings, Vector Databases, Model Evaluation (RAGAS), LLMOps, Model Context Protocol (MCP)

APIs & Models: Anthropic Claude API, OpenAI API, Google Gemini API, Hugging Face Transformers

Frameworks & Tools: LangChain, LangGraph, LangSmith, CrewAI, AutoGen, FastAPI, Streamlit, Next.js, React, n8n

Languages: Python, TypeScript, JavaScript, SQL

Cloud & MLOps: GCP (Cloud Run, Cloud SQL, BigQuery, Cloud Scheduler, Secret Manager, Workload Identity Federation, GCS), AWS (Fargate, ECR), Docker, GitHub Actions, CI/CD, Vercel

ML & Data: NLP, Deep Learning, PyTorch, TensorFlow, Scikit-learn, Computer Vision, Statistics, PostgreSQL, SQLite, BigQuery, Vector Stores

Integrations: HubSpot API, Google Slides API, Google Drive API, SAM.gov, HigherGov, Linear

PROFESSIONAL EXPERIENCE

AI Engineer - FEDCON, Tampa (USA)

Feb 2026 – Present

- Architected and own an LLM-powered opportunity-scoring engine (Claude API) that ranks federal solicitations against client profiles across 19 weighted dimensions and feeds 7 downstream consumers across 4 repos; cut inference to ~\$0.01 per prospect via prompt caching and Haiku/Sonnet model-tiering at ~190-client scale.
- Rebuilt the SAM.gov data pipeline on GCP, splitting 2.3M records into active/archive tables with daily, weekly, and monthly Cloud Run jobs, and lifted capture rate to 95%+ by fixing a 35% data-loss bug reducing 250+ paginated API calls to 2–3 via SAM's async Extract API.
- Reduced a core production prompt from 440K to ~50K characters (~88%) by re-architecting the client-assumptions generator with topic-aware pre-processing (OVERWRITE / ACCUMULATE / SUPERSEDE patterns over 156+ log entries), eliminating stale-data regressions across 180 clients.
- Shipped an end-to-end prospect-deck automation pipeline (HubSpot → USASpending / HigherGov market-intel synthesis → Gemini 12-slide generation → Google Slides API), delivering the first auto-generated deck for a \$319M-market client in April 2026.
- Designed a two-model LLM insights-extraction architecture (Claude Haiku 4.5 per-entry extraction → SQLite store → Sonnet 4.6 format pass), measured at \$0.39 per refresh and ~\$310/year to operate at 190-client scale.
- Productized a departing engineer's local Python opportunity scanner into a cloud app (Next.js 16, TypeScript, Drizzle, Cloud SQL, Vercel) with a 6-factor scoring model, auth, caching, and automated 2 AM ET daily scans.

Research Assistant - University of South Florida, Tampa, FL

Aug 2024 – May 2025

AI & Data Extraction for Scientific Literature

- Spearheaded adaptation of the EXSCLAIM! tool to autonomously extract microscopy images from scientific literature, building a scalable protein-polymer image repository on the Circe HPC (high-performance computing) cluster for ML research.
- Extended the journal-scraping pipeline beyond Nature to RSC's Soft Matter, troubleshooting Selenium WebDriver and optimizing HTML-parsing strategies for reliable large-scale extraction.
- Administered a private GitHub repository with SSH authentication and Circe HPC execution, ensuring structured image storage and scalable retrieval for downstream ML.

Machine Learning Intern - Suvidha Foundation, India

Dec 2021 – Mar 2022

- Surveyed 100–150 NLP research papers, documenting methodologies, limitations, and trade-offs across Transformer, RNN, and encoder–decoder architectures.
- Compared semantic-graph and sequence-to-sequence models for summarization and text generation, compiling a structured knowledge base to guide future model selection.

PROJECTS

- Multi-Agent Stock Prediction Agent:** Python, LangChain, LangGraph, MCP
- Orchestrated a multi-agent system (LangChain, LangGraph, MCP / Bright Data) coordinating stock-discovery, market-data, and news-sentiment agents under a supervisor to produce buy/sell/hold calls with target entry prices for NYSE-listed companies.
 - Produced sentiment-backed recommendations and target entry prices that improved the interpretability of AI-driven trading research.
- Document Analysis, Chat & Comparison Portal with Advanced RAG:** Python, LangChain, FastAPI, AWS
- Engineered and deployed an advanced RAG pipeline (ingestion → retrieval → generation) in LangChain with custom chunking/indexing, lifting retrieval accuracy via query rewriting (RAG Fusion / ReAct) and MMR reranking.
 - Delivered the full stack end-to-end — FastAPI backend, Streamlit UI — with automated CI/CD (GitHub Actions, Docker) for scalable deployment on AWS Fargate and ECR with Secrets Manager.
 - Optimized throughput with local LLMs (vLLM / Groq) and quantized models, and implemented Cache-Augmented Generation (CAG) to cut latency on repeated queries for production-ready, low-latency serving.
- Bone Fracture Detection (CNN):** Python, TensorFlow, Deep Learning
- Trained a convolutional neural network on the MURA v1.1 X-ray dataset to detect fractures, achieving 87.5% accuracy, 92% precision, 94.29% specificity, and an 85.18% F1 score.
 - Streamlined large medical-imaging data handling and optimized computational efficiency for clinical-scale inference.
- Flight Price Prediction:** Python, Scikit-learn, Pandas
- Cleaned and engineered a 10,000+-row, 41-feature flight dataset and trained regression models (Random Forest, Decision Tree, KNN), reaching 95–97% training accuracy.
 - Benchmarked the models against linear regression and selected the strongest by validation performance, with full preprocessing and feature-engineering pipelines.

PUBLICATIONS

- *Abstractive Text Summarization using a fine-tuned PEGASUS model* — JETIR (Journal of Emerging Technologies and Innovative Research), 2023; fine-tuned and benchmarked PEGASUS against BERT, T5, XLNet, and GPT-2 on the SAMSum dataset, achieving strong ROUGE scores.

EDUCATION

M.S., Data Intelligence — University of South Florida, Tampa, FL GPA: 3.8/4.0	Jan 2024 – Dec 2025
B.Tech., Electronics & Telecom Engineering — University of Mumbai, India GPA: 8.9/10	Jul 2019 – Jul 2023